Experiment/ Case Study No.: 4

1. Experiment Title: Use function types, scope, and closures; apply try-catch; build a palindrome checker.

2. Software/Tools Required:

- Visual Studio Code (VS Code) / Notepad++
- Google Chrome / Mozilla Firefox
- JavaScript (ES6)
- HTML5
- Windows/Linux Operating System

3. Experiment Program Code:

1. practical4.html:

```
<!DOCTYPE html>
<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Palindrome Analyzer</title>

<link rel="stylesheet" href="practical4.css">

<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&displ
ay=swap" rel="stylesheet">

</head>

<body>

<div class="container">

<h1>🔤 Palindrome Analyzer</h1>
```

```
    <p>Check whether a word or sentence is a palindrome.</p>

    <input type="text" id="textInput" placeholder="Enter text">

    <div class="buttons">

    <button onclick="checkPalindrome()">Check</button>

    <button onclick="clearData()">Reset</button>

    </div>

    <div class="result">

    <h2 id="result">Result will appear here</h2>

    <p id="message"></p>

    <p id="count"></p>

    </div>

    </div>

    <script src="practical4.js"></script>

    </body>

    </html>
```

2. practical4.css

```
/* ==============================
  Palindrome Analyzer
  style.css
============================== */

*{
   margin:0;
   padding:0;
   box-sizing:border-box;
   font-family:'Poppins',sans-serif;
}

body{

   display:flex;
   justify-content:center;
   align-items:center;
   height:100vh;
```

```
    background:linear-gradient(135deg,#4facfe,#00f2fe);
}
.container{
    width:420px;
    background:#fff;
    padding:30px;
    border-radius:15px;
    text-align:center;
    box-shadow:0 10px 25px rgba(0,0,0,0.2);
}
h1{
    color:#0077b6;
    margin-bottom:10px;
}
p{
    color:#555;
    margin-bottom:20px;
}
input{
    width:100%;
    padding:12px;
    border:2px solid #0077b6;
    border-radius:8px;
    outline:none;
    font-size:16px;
```

```
}
input:focus{
    border-color:#00b4d8;
    box-shadow:0 0 10px rgba(0,180,216,.3);
}
.buttons{
    display:flex;
    justify-content:space-between;
    margin-top:20px;
}
button{
    width:48%;
    padding:12px;
    border:none;
    border-radius:8px;
    font-size:16px;
    font-weight:bold;
    cursor:pointer;
    transition:.3s;
}
button:first-child{
    background:#0077b6;
    color:white;
}
button:last-child{
```

```
    background:#ef233c;
    color:white;
}
button:hover{
    transform:scale(1.05);
}
.result{
    margin-top:25px;
    background:#f4f4f4;
    padding:20px;
    border-radius:10px;
}
#result{
    color:#023e8a;
    margin-bottom:10px;
}
#message{
    font-size:16px;
    color:#333;
}
#count{
    margin-top:10px;
    font-weight:bold;
    color:#009688;
}
```

```
/* Responsive */

@media(max-width:500px){

.container{

width:90%;

}

.buttons{

flex-direction:column;

gap:10px;

}

button{

width:100%;

}

}
```

3. practical4.js:

```
// ========================================
// Palindrome Analyzer
// script.js
// ========================================

// Variable Scope
var appName = "Palindrome Analyzer";
const version = "1.0";

// Closure Function
function visitCounter() {

    let count = 0;

    return function () {

        count++;

        return count;

    };
```

```
}

const counter = visitCounter();

// Function Declaration
function cleanText(text) {

   return text.toLowerCase().replace(/[^a-z0-9]/g, "");

}

// Function Expression
const reverseString = function(text) {

   return text.split("").reverse().join("");

};

// Arrow Function
const isPalindrome = (text) => {

   return text === reverseString(text);

};

// Main Function
function checkPalindrome() {

   try {

      let input = document.getElementById("textInput").value.trim();

      if (input === "") {

         throw "Please enter a word or sentence.";

      }

      let cleaned = cleanText(input);

      if (cleaned.length === 0) {

         throw "Input contains only special characters.";

      }

      let result = isPalindrome(cleaned);

      let message = document.getElementById("message");
```

```
        let heading = document.getElementById("result");

        let count = document.getElementById("count");

        if (result) {

            heading.innerHTML = "☑ Palindrome";

            message.innerHTML =
            `"${input}" reads the same forward and backward.`;

            heading.style.color = "green";

        }

        else {

            heading.innerHTML = "❌ Not a Palindrome";

            message.innerHTML =
            `"${input}" is not a palindrome.`;

            heading.style.color = "red";

        }

        count.innerHTML =
        "Characters Entered : " + cleaned.length;

        console.log(`Checked : ${input}`);

        console.log(`Total Checks : ${counter()}`);

    }

    catch(error) {

        document.getElementById("result").innerHTML = "⚠ Error";

        document.getElementById("message").innerHTML = error;

        document.getElementById("count").innerHTML = "";

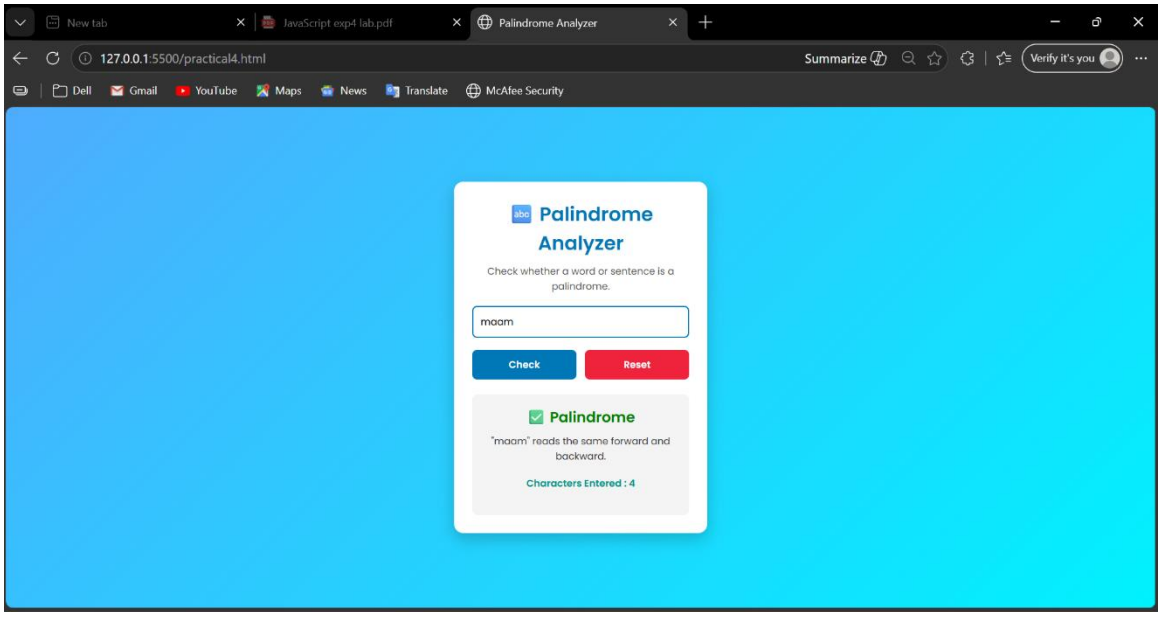
        document.getElementById("result").style.color = "orange";

    }

}
```

```
// Anonymous Function
document.getElementById("textInput").addEventListener("keypress",
function(event){
    if(event.key==="Enter"){
        checkPalindrome();
    }
});
// Reset
function clearData(){
    document.getElementById("textInput").value="";
    document.getElementById("result").innerHTML="Result will appear here";
    document.getElementById("message").innerHTML="";
    document.getElementById("count").innerHTML="";
    document.getElementById("result").style.color="#023e8a";
}
```

4. Output:



Palindrome Analyzer

5. Case Study Title: **Password Strength Checker**

6. Case Study Program Code:

a. caseStudy4.html:

```
<!DOCTYPE html>
<html>

<head>

<title>Password Strength Checker</title>

<link rel="stylesheet" href="caseStudy4.css">

</head>

<body>
```

```
<div class="container">

<h1>Password Strength Checker</h1>

<input type="password" id="password" placeholder="Enter Password">

<br><br>

<button onclick="checkPassword()">

Check Password

</button>

<button onclick="resetData()">

Reset

</button>

<h2 id="result"></h2>

<p id="message"></p>

</div>

<script src="caseStudy4.js"></script>

</body>

</html>
```

b. script.js:

```
/* ==================================
  Password Strength Checker
  style.css
================================== */

*{
   margin:0;
   padding:0;
   box-sizing:border-box;
   font-family:'Poppins',sans-serif;
```

```
}
body{
    display:flex;
    justify-content:center;
    align-items:center;
    height:100vh;
    background:linear-gradient(135deg,#4e54c8,#8f94fb);
}
.container{
    width:420px;
    background:white;
    padding:30px;
    border-radius:15px;
    text-align:center;
    box-shadow:0 10px 25px rgba(0,0,0,0.3);
}
h1{
    color:#4e54c8;
    margin-bottom:20px;
}
input{
    width:100%;
    padding:12px;
    font-size:16px;
    border:2px solid #4e54c8;
```

```
    border-radius:8px;
    outline:none;
    margin-bottom:20px;
    transition:0.3s;
}
input:focus{
    border-color:#6c63ff;
    box-shadow:0 0 10px rgba(108,99,255,.4);
}
button{
    width:45%;
    padding:12px;
    margin:5px;
    border:none;
    border-radius:8px;
    font-size:16px;
    cursor:pointer;
    font-weight:bold;
    transition:.3s;
}
button:first-child{
    background:#4CAF50;
    color:white;
}
button:last-child{
```

```
    background:#f44336;
    color:white;
}
button:hover{
    transform:scale(1.05);
}
#result{
    margin-top:20px;
    font-size:28px;
}
#message{
    margin-top:15px;
    font-size:17px;
    color:#444;
    font-weight:500;
}
/* Password Strength Meter */
.strength{
    width:100%;
    height:18px;
    background:#ddd;
    border-radius:20px;
    overflow:hidden;
    margin-top:20px;
}
```

```
#progress{
   width:0%;
   height:100%;
   background:green;
   transition:1s;
}
/* Responsive */
@media(max-width:500px){
.container{
width:90%;
}
button{
width:100%;
margin-top:10px;
}
}
```

c. caseStudy4.js:

```
// ========================================
// Password Strength Checker
// script.js
// ========================================

// Variable Scope
var application = "Password Strength Checker";
const MIN_LENGTH = 8;

// Closure Function
function checkCounter(){
   let count = 0;
```

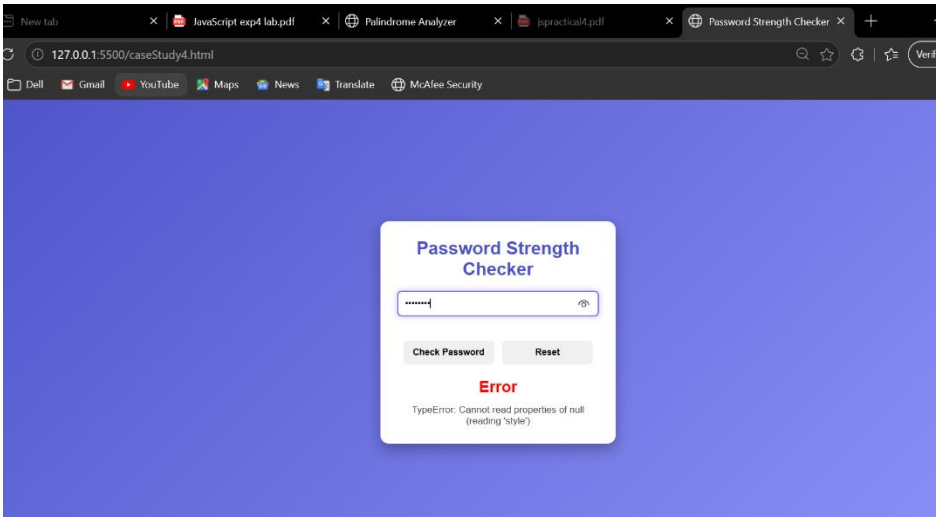
```
    return function(){
        count++;
        return count;
    }
}
const totalChecks = checkCounter();
// Function Declaration
function validatePassword(password){
    if(password.length < MIN_LENGTH){
        throw "Password must contain at least 8 characters.";
    }
}
// Function Expression
const calculateStrength = function(password){
    let score = 0;
    if(/[A-Z]/.test(password)) score++;
    if(/[a-z]/.test(password)) score++;
    if(/[0-9]/.test(password)) score++;
    if(/[!@#$%^&*()_+?><]/.test(password)) score++;
    if(password.length >= 8) score++;
    return score;
};
// Arrow Function
const getStrength = (score)=>{
    if(score<=2)
```

```
        return "Weak";
    else if(score==3 || score==4)
        return "Medium";
    else
        return "Strong";
};
// Main Function
function checkPassword(){
    try{
        let password = document.getElementById("password").value.trim();
        validatePassword(password);
        let score = calculateStrength(password);
        let strength = getStrength(score);
        let result = document.getElementById("result");
        let message = document.getElementById("message");
        let progress = document.getElementById("progress");
        result.innerHTML = strength + " Password";
        if(strength=="Weak"){
            result.style.color="red";
            message.innerHTML="Your password is weak. Add uppercase letters,
numbers and special characters.";
            progress.style.width="35%";
            progress.style.background="red";
        }
        else if(strength=="Medium"){
            result.style.color="orange";
```

```
            message.innerHTML="Good password. Add more security features.";
            progress.style.width="70%";
            progress.style.background="orange";
        }
        else{
            result.style.color="green";
            message.innerHTML="Excellent! Your password is secure.";
            progress.style.width="100%";
            progress.style.background="green";
        }
        console.log(`Password Checked Successfully`);
        console.log(`Total Checks : ${totalChecks()}`);
    }
    catch(error){
        document.getElementById("result").innerHTML="Error";
        document.getElementById("result").style.color="red";
        document.getElementById("message").innerHTML=error;
        document.getElementById("progress").style.width="0%";
    }
}
// Anonymous Function
document.getElementById("password").addEventListener("keypress",
function(event){
    if(event.key==="Enter"){
        checkPassword();
```

```
    }
});
// Reset Function
function resetData(){
    document.getElementById("password").value="";
    document.getElementById("result").innerHTML="";
    document.getElementById("message").innerHTML="";
    document.getElementById("progress").style.width="0%";
}
```

7. Output:



[Password Strength Checker]

8. Result/Conclusion: The project was successfully developed using **HTML, CSS, and JavaScript**. It demonstrates the implementation of **different JavaScript function types, variable scope, closures, exception handling, and input validation** to process user input accurately. The application provides an interactive and user-friendly interface while improving logical programming skills and reducing manual errors. This practical successfully fulfills the objectives of implementing JavaScript concepts in a real-world web application and can be further enhanced with additional security features and database integration.